Adjacency matrices, walks, and connectivity: a study note

November 2, 2026

Abstract

A graph with N vertices can be described by an N by N table of zeros and ones, its adjacency matrix A. This note answers three questions about that table. First, the entry in row i and column j of the k-th power of A is the number of walks of length exactly k from vertex n_i to vertex n_j; it counts walks, not paths, and the note shows by example why the difference matters. Second, a graph with $N \geq 2$ vertices is connected exactly when every off-diagonal entry of $A + A^2 + \cdots + A^{N-1}$ is positive, which is the same as saying that $(I + A)^{N-1}$ has no zero entry; the note proves this and shows that the last power cannot be lowered. Third, a path between n_i and n_j, in the convention of the course, is a walk from n_i to n_j in which no vertex occurs twice; the note compares this with the textbook's two words for related ideas. Definitions taken from the course textbook carry the textbook's numbers, every statement is proved or attributed, and every matrix in the worked examples was computed by a script.

1 Introduction

1.1 The question and where it came from

A graph is a set of points, called vertices, together with a set of lines, called edges, each joining two of the points. A graph with N vertices can be recorded as an N by N table of zeros and ones called the adjacency matrix A, which has a one in row i and column j when the vertices numbered i and j are joined by an edge. This note studies what such a table can tell us beyond the obvious. The study has three questions.

The first two questions are written on a lecture whiteboard, photographed twice from slightly different angles [4, 5]. The board states a problem: a graph with N nodes is given through its adjacency matrix A. Question 1 asks what an entry of a power of A represents. Question 2 asks how to know from A whether the graph is connected. Below the questions the board defines a graph to be connected when for any two nodes n_i, n_j there exists a path between the nodes [4, 5]. The third question is added by the assignment text that accompanied the photographs; it asks for the definition of a path between n_i and n_j. The assignment text also fixes the setting: the graph is finite, simple, and undirected, and "path" means a walk in which no vertex is repeated. The assignment text is the source of these two assumptions, and nothing on the board contradicts them. Appendix C transcribes the board line by line.

Q1. What does the entry $(A^k)_{ij}$ represent?

Q2. How can one determine from A whether the graph is connected?

Q3. What is the definition of a path between the vertices n_i and n_j?

The mark above A. In question 1 the board has a small mark above the letter A. In both photographs it looks like a small n or a pair of quotation marks, not clearly a k. The assignment text asks about A^k, so this note reads the mark as a positive whole number k (the power). This is an interpretation, not something the photographs settle. The answer below does not depend on it, because the theorem of Section 3 holds for every positive whole number used as the exponent.

The example on the board. The board shows a drawing of three vertices labelled 1, 2, 3 beside a matrix with rows 0 1 0, 1 0 1, 0 1 0 [4, 5]. The matrix says that vertex 1 is joined to vertex 2 and vertex 2 to vertex 3, and that vertex 1 is not joined to vertex 3. The drawing shows three lines: from 1 to 2 (with an arrowhead at 2), from 1 to 3, and from 2 to 3. So the drawing has an edge from 1 to 3 that the matrix does not have. The problem statement says the graph is given through its adjacency matrix, so this note takes the matrix to define the example graph, called B below, and reports the mismatch rather than guessing which the lecturer meant. The matrix is symmetric, so it does not support the arrowhead, which would mean a one-way edge; the arrowhead is treated as a stray mark.

1.2 The story used throughout

Every definition, theorem, test, and worked example in this note is introduced first by a short paragraph headed "In plain words". Those paragraphs tell one story. A village has houses. Each road joins exactly two different houses. No two roads join the same pair of houses. Every road can be driven in both directions. A trip is a list of houses, one after another, where each next house is at the far end of a road from the house before. A trip may pass a house twice, and it may drive a road twice. A route is a trip that never passes the same house twice. Table 1 shows how the story words match the mathematical words that are defined in Section 2.

Table 1: The story words and the mathematical words they stand for.

Story word	Mathematical word
house	vertex
road	edge
trip	walk
trip that never drives a road twice	trail
route (trip that never passes a house twice)	path
village in one piece	connected graph
piece of a village	component

The running example is the village of three houses in a row: house 1, house 2, house 3, a road from 1 to 2 and a road from 2 to 3, and no road from 1 to 3. This is the graph B of the board matrix.

1.3 Conventions and notation

The course textbook is Petrosyan, Mkrtchyan, and Kamalyan, *Graph Theory* [9]. Its definitions are cited by their own numbers, written "textbook Definition 1.2.10" to keep them apart from the numbers of this note. Page numbers of the textbook are printed page numbers. Letters are fixed once, here.

- A graph is $G = (V, E)$ with a finite nonempty vertex set V and an edge set E of unordered pairs of distinct vertices [9, Section 1.1, Definition 1.1.1]. The textbook names the vertices $v_1, \dots, v_n$. The board names them $n_1, \dots, n_N$ and writes (G, V) and "$e_{ij} \in V$" where the textbook writes $G = (V, E)$ and $e \in E$ [4, 5]. This is a difference of notation only. This

note uses the board's vertex names $n_1, \ldots, n_N$, the textbook's V and E, and N for the number of vertices.

- An edge joining n_i and n_j is written $n_i n_j$, or e_{ij} as on the board. In walk lists the vertex n_i is written by its index, so 1-2-3 means n_1, n_2, n_3.
- A is the adjacency matrix and A_{ij} its entry in row i and column j; A^k is the k-th power of A and $(A^k)_{ij}$ its (i, j) entry; I is the N by N identity matrix, with ones on the diagonal and zeros elsewhere.
- $\deg(n_i)$ is the degree of n_i and $d(u, v)$ the distance between u and v (Section 2).
- The word *path* is used in the course sense: no repeated vertex. The textbook uses the word "path" for something weaker, which this note calls a *trail*; Section 5 compares the conventions in one table.

1.4 What this note shows

The textbook defines graphs, walks, paths, and connectivity, but, according to a full-text search of the whole book for the adjacency matrix, it contains no theorem on the powers of A and no matrix test for connectivity; the book was not read page by page outside the chapters used here, so this absence is reported with that qualification [9]. Both results are therefore taken from other sources and proved in full in this note. The results are these.

- **Answer to Q1** (Theorem 3.1). $(A^k)_{ij}$ is the number of walks of length k from n_i to n_j [2, 3, 6, 1]. It is a count of walks and not of paths (Proposition 3.6); one published statement that says "paths" is wrong as written [7].
- **Answer to Q2** (Theorem 4.7). For $N \geq 2$, G is connected if and only if every off-diagonal entry of $A + A^2 + \cdots + A^{N-1}$ is positive, and also if and only if every entry of $(I + A)^{N-1}$ is positive [7, 1]. These sources state the test; the proof here is the note's own.
- **Answer to Q3** (Definition 2.34 and Section 5). A path between n_i and n_j is a walk from n_i to n_j with no repeated vertex, the textbook's simple path (Definition 1.2.13) [9].

1.5 Organization

Section 2 builds the vocabulary in the textbook's order: graph, adjacency matrix, walk, path, cycle, distance, connectivity. Section 3 answers Q1, Section 4 answers Q2, and Section 5 answers Q3 and compares the naming conventions. Section 6 works the three-vertex examples through with every matrix written out. Section 7 states the limits of the results, and Section 8 gives the direct answers. The appendices hold secondary proofs, the textbook locators, the transcription of the board, and the way to rerun the computations.

2 Background: graphs, matrices, walks, and paths

This section builds the vocabulary in the order of the textbook [9]: graph, adjacent vertices, adjacency matrix, degree, walk, closed walk, trail, path, cycle, distance, and connectivity. Nothing is used before it is defined. Every definition is followed by a remark on what it allows and forbids, an example, and a non-example. A definition that comes from the textbook carries the textbook's number. A proposition, lemma, or corollary whose proof is printed here is this note's own statement of the result; a citation in its first sentence names a source that also states it. Four small graphs are used for the examples.

- B is the board graph read from the board matrix: vertices n_1, n_2, n_3 and edges n_1n_2 and n_2n_3.
- D has vertices n_1, n_2, n_3 and the single edge n_1n_2; it is the disconnected example.
- T is the board drawing read as drawn: vertices n_1, n_2, n_3 and edges n_1n_2, n_1n_3, n_2n_3, a triangle. It is used only where a graph with a cycle is needed, and it is not the running example.
- H has vertices $v_1, \ldots, v_5$ and the seven edges v_1v_2, v_1v_3, v_2v_3, v_2v_4, v_3v_4, v_3v_5, v_4v_5. These are the edges that the textbook's own examples of walks, paths, and cycles use [9, p. 22]; H is used to show differences that B is too small to show.

Figure 1 draws B, D, T, and H.

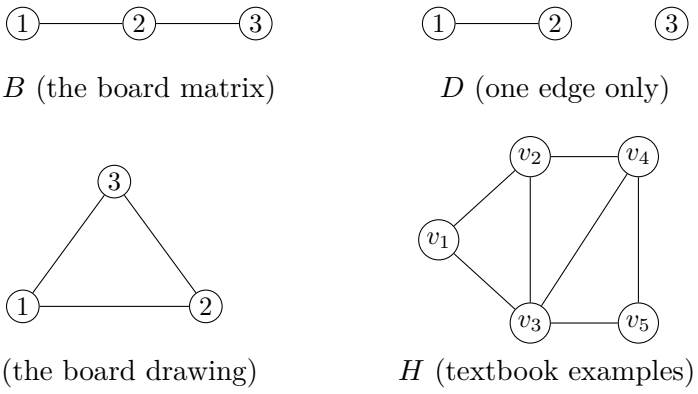


Figure 1: The four graphs used for examples. B is the running example and D is the disconnected example. T is the triangle in the board drawing. H has the seven edges used by the textbook's examples of walks, paths, and cycles.

2.1 Graphs, adjacency, and the adjacency matrix

In plain words. A village has a list of houses and a list of roads. A road always joins two different houses. No two roads join the same pair of houses. A road can be driven both ways.

Definition 2.1 (simple undirected graph; textbook Definition 1.1.1)**.** A *simple undirected graph* is a pair $G = (V, E)$ where V is a finite nonempty set of *vertices* and E is a set of unordered pairs of distinct vertices, the *edges* [9, Section 1.1].

Remark 2.2. The definition forbids a loop (an edge from a vertex to itself, since the two ends of an edge must be distinct) and a repeated edge (since E is a set, a pair occurs at most once). It forbids direction, since the pair is unordered and n_in_j and n_jn_i are the same edge. It allows vertices that belong to no edge.

Example 2.3. B is a simple undirected graph with $V = \{n_1, n_2, n_3\}$ and $E = \{n_1n_2, n_2n_3\}$.

Non-example 2.4. Two vertices joined by two roads repeat an edge, and a road from n_1 to n_1 is a loop; neither is a simple graph.

In plain words. Two houses are neighbours when one road joins them directly. Being neighbours does not say anything about longer trips.

Definition 2.5 (adjacent vertices; textbook Definition 1.1.4)**.** Vertices u and v are *adjacent* if $uv \in E$; then u is a *neighbour* of v [9, p. 11].

Remark 2.6. Adjacency is symmetric, because $uv = vu$, and no vertex is adjacent to itself, because there are no loops. Adjacency says nothing about connections through other vertices.

Example 2.7. In B the vertices n_1 and n_2 are adjacent.

Non-example 2.8. In B the vertices n_1 and n_3 are not adjacent, although a trip of two roads joins them.

In plain words. Make a square table with one row and one column for every house. Write a one where the row house and the column house share a road. Write a zero everywhere else. The table describes the whole village.

Definition 2.9 (adjacency matrix; textbook, after Definition 1.1.6, p. 11)**.** Let $G = (V, E)$ with $V = \{n_1, \ldots, n_N\}$. The *adjacency matrix* of G is the N by N matrix A whose entry in row i and column j is $A_{ij} = 1$ if n_i and n_j are adjacent and $A_{ij} = 0$ otherwise [9, p. 11].

The textbook gives this definition without a number of its own, right after Definition 1.1.6. The board writes the same rule as $A_{ij} = 1$ if e_{ij} belongs to the edge set and $A_{ij} = 0$ if it does not [4, 5].

Remark 2.10. The definition allows only the entries 0 and 1. It depends on how the vertices are numbered: renumbering permutes the rows and the columns together.

Example 2.11. For B the adjacency matrix is $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$, which is the matrix on the board: $A_{12} = A_{21} = 1$, $A_{23} = A_{32} = 1$, and every other entry is 0.

Non-example 2.12. The matrix with rows $(0\,1\,0)$, $(0\,0\,1)$, $(0\,1\,0)$ is not the adjacency matrix of any simple undirected graph, because $A_{12} = 1$ but $A_{21} = 0$, which would need a one-way road. A matrix with a one on the diagonal would need a loop, and one with an entry 2 would need a double road.

In plain words. Later we multiply this table by itself. We say here exactly what that means. To fill one place of the new table, take a row of the first table and a column of the second. Multiply the numbers that sit in the same position, and add the products.

Definition 2.13 (matrix product and power)**.** For N by N matrices X and Y the *product* XY is the N by N matrix with entries $(XY)_{ij} = \sum_{r=1}^{N} X_{ir} Y_{rj}$. The *powers* of A are $A^1 = A$ and $A^k = A^{k-1}A$ for $k \geq 2$; the *identity matrix* I satisfies $IX = XI = X$ and is used as $A^0 = I$ only where stated.

Remark 2.14. The product is row times column, and it is a sum of N terms. It is not commutative in general, but powers of one matrix A commute with each other.

Example 2.15. For B, $(A^2)_{11} = 0 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 = 1$.

Non-example 2.16. Squaring each entry of A separately gives A again, because its entries are 0 and 1; this is not A^2.

In plain words. The number of roads at a house tells how busy it is. The table lets us read that number straight off. Adding one row of the table gives it.

Definition 2.17 (degree; textbook Definition 1.2.1)**.** The *degree* $\deg(n_i)$ of a vertex n_i is the number of edges that contain n_i [9, p. 16].

The textbook writes $d_G(v)$ or $d(v)$ for the degree and defines it after the adjacency matrix, which is why it appears here after Definition 2.9.

Remark 2.18. In a simple graph the degree is also the number of neighbours, since each neighbour is joined to n_i by exactly one edge.

Example 2.19. In B, $\deg(n_1) = 1$, $\deg(n_2) = 2$, $\deg(n_3) = 1$.

Non-example 2.20. $\deg(n_1)$ is not 2 in B, even though two other vertices can be reached from n_1: only edges at n_1 count, and n_3 is reached by going through n_2.

In plain words. The table has easy features. It looks the same when flipped over its diagonal. The diagonal holds only zeros. Adding up one row counts the roads at that house.

Proposition 2.21 (properties of A; textbook, pp. 11–12)**.** *Let A be the adjacency matrix of a simple undirected graph with N vertices and $|E|$ edges.*

1. *Every entry of A is* 0 *or* 1*.*
2. *A is symmetric: $A_{ij} = A_{ji}$ for all i, j.*
3. *The diagonal is zero: $A_{ii} = 0$ for all i.*
4. *The sum of row i, and of column i, equals* $\deg(n_i)$*.*
5. *The sum of all entries equals* $2|E|$*.*
6. *Conversely, every N by N matrix of zeros and ones that is symmetric with zero diagonal is the adjacency matrix of a simple undirected graph on N numbered vertices.*

Items 1 to 3 and 6 are stated in the textbook [9, pp. 11–12], item 6 is also stated by Paramadevan and Sotheeswaran [7], and item 5 is the textbook's Theorem 1.2.1 (Euler) [9, p. 16]. Item 4 is proved here; Paramadevan and Sotheeswaran state a version of it that allows loops [7].

Proof. Items 1 to 3 are read off Definition 2.9 and Definition 2.5: adjacency is symmetric and no vertex is adjacent to itself. Item 4: the sum of row i is the number of j with $A_{ij} = 1$, which is the number of neighbours of n_i. Each neighbour is joined to n_i by exactly one edge, and no edge at n_i joins it to itself, so this number is the number of edges at n_i, that is $\deg(n_i)$. Column i is the same by symmetry. Item 5: add item 4 over all i; every edge contributes one to the degree of each of its two ends, so it is counted twice. Item 6: join n_i and n_j exactly when the entry is 1; symmetry makes this a set of unordered pairs, and the zero diagonal excludes loops. □

For the board matrix the row sums are $1, 2, 1$, the column sums are $1, 2, 1$, and the total is $4 = 2|E|$ with $|E| = 2$; these were checked by script (Appendix D).

2.2 Walks, trails, paths, and cycles

In plain words. A trip is a list of houses, one after another. Each next house must sit at the far end of a road from the house before. You may pass a house twice. You may drive a road twice, and then it counts twice. The length of a trip is the number of roads you drove. Staying at home is also a trip, of length zero.

Definition 2.22 (walk; textbook Definition 1.2.10)**.** A *walk of length k from u_0 to u_k*, also called a (u_0, u_k)-walk, is a sequence $u_0, e_1, u_1, \ldots, u_{k-1}, e_k, u_k$ of vertices and edges with $e_i = u_{i-1}u_i$ for $1 \le i \le k$ [9, p. 21]. The *length* of the walk is k, the number of edges, each edge counted as often as it occurs [9, p. 22].

Remark 2.23. Vertices and edges of a walk may repeat [9, p. 22]. The definition forbids only a jump: two consecutive vertices that are not joined by an edge. A walk of length k lists $k+1$ vertices. In a simple graph there is at most one edge between two vertices, so the edge e_i is fixed by the vertices u_{i-1} and u_i. Hence a walk is the same thing as its vertex sequence $u_0, u_1, \ldots, u_k$ in which every two consecutive vertices are adjacent, and two walks differ exactly when their vertex sequences differ at some position. The proofs below count such sequences. The textbook does not say whether $k = 0$ is allowed. This note allows it, as an interpretation, because the textbook's own property $d(u, u) = 0$ of the distance (Definition 2.43) needs a path of length 0 from u to u [9, p. 23].

Example 2.24. In B the sequence $1, 2, 1, 2, 3$ is a walk from n_1 to n_3 of length 4: its steps $12, 21, 12, 23$ are all edges. The textbook's example in H is the walk $v_1, v_3, v_2, v_4, v_3, v_2, v_4, v_3, v_5$ of length 8 [9, p. 22].

Non-example 2.25. In B the sequence $1, 3$ is not a walk, because n_1n_3 is not an edge of B. It would be a walk in T. The textbook's non-example is $v_3, v_1, v_4, v_2, v_3, v_5$, which is not a walk of H because v_1v_4 is not an edge [9, p. 22].

In plain words. A trip is closed when it ends at the house where it began. That is the only new rule.

Definition 2.26 (closed walk; textbook Definition 1.2.11)**.** A (u_0, u_k)-walk is *closed* if $u_0 = u_k$ [9, p. 22].

Remark 2.27. A closed walk may still repeat vertices and edges. In a simple graph no closed walk has length 1, since that would be a loop, and a closed walk of length 2 is u, w, u, which drives one edge twice.

Example 2.28. In B the walk $2, 1, 2$ is closed, of length 2.

Non-example 2.29. In B the walk $1, 2, 3$ is not closed: it starts at n_1 and ends at n_3.

In plain words. Now add one rule. Never drive the same road twice. You may still pass a house twice, if you came back by a different road. English books call this kind of trip a trail. Our textbook calls it a path. This note says trail, so that the word path keeps one meaning.

Definition 2.30 (trail; textbook Definition 1.2.12, where it is called a "path")**.** A (u_0, u_k)-walk is a *trail* if its edges $u_0u_1, \ldots, u_{k-1}u_k$ are pairwise distinct [9, p. 22].

The textbook's word for this is "path". In English usage a walk with distinct edges is a trail [3]. To point at the textbook's word, this note writes "textbook path".

Remark 2.31. A trail allows repeated vertices and forbids repeated edges. A vertex can recur in a trail only if the trip left it by one road and came back by another.

Example 2.32. In H the sequence $v_1, v_3, v_2, v_4, v_3, v_5$ is a trail: its edges $v_1v_3, v_3v_2, v_2v_4, v_4v_3, v_3v_5$ are distinct, although v_3 occurs twice [9, p. 22].

Non-example 2.33. In H the length-8 walk of the example after Definition 2.22 is not a trail, because the edge v_3v_2 occurs twice [9, p. 22]. In B the walk $1, 2, 1, 2, 3$ is not a trail, because the edge n_1n_2 occurs twice.

In plain words. Now add a stronger rule. Never pass the same house twice. Then you also never drive the same road twice. A trip like this goes straight from start to end without any loop. This is what our course calls a path.

Definition 2.34 (path; textbook Definition 1.2.13, where it is called a "simple path")**.** A (u_0, u_k)-walk is a *path* if its vertices $u_0, \ldots, u_k$ are pairwise distinct [9, p. 22]. When $u_0 = u$ and $u_k = v$ it is a *path from u to v*, or a *path between u and v*.

The textbook calls this a simple path and defines it as a path in its own sense (a trail) with distinct vertices [9, p. 22]. Proposition 2.38 below shows that distinct vertices already give distinct edges, so the two ways of saying it agree. In English usage a path is a walk with distinct vertices [3, 8]. That "path" means this in the course is stated by the assignment text and is not printed in the textbook; the board does not say which meaning of "path" it uses [4, 5]. This note therefore uses Definition 2.34 for "path" everywhere, as an interpretation of the course convention, and Section 5 compares the conventions.

Remark 2.35. A path forbids repeated vertices and therefore repeated edges (Proposition 2.38). It allows length 0, a single vertex, and length 1, a single edge. It does not allow a closed path of positive length, since the first and last vertex would be equal.

Example 2.36. In B the sequence $1, 2, 3$ is a path from n_1 to n_3 of length 2, and it is the only one. In H the sequence v_1, v_3, v_4, v_5 is a path [9, p. 22].

Non-example 2.37. In B the sequence $1, 2, 3, 2$ is a walk but not a path, because n_2 occurs twice. In H the trail $v_1, v_3, v_2, v_4, v_3, v_5$ is not a path, because v_3 occurs twice [9, p. 22].

In plain words. If you never pass a house twice, you never drive a road twice either. The reason is short. Driving a road twice means standing at both of its ends twice. So every route is also a trip without repeated roads.

Proposition 2.38 (a path is a trail)**.** *In a simple graph, a walk whose vertices are pairwise distinct has pairwise distinct edges. Hence every path is a trail, and every trail is a walk.*

Grinberg states this as an exercise [3]; the proof is given here.

Proof. Let $u_0, \ldots, u_k$ have distinct vertices, and take two steps $a < b$, that is, the edges $u_{a-1}u_a$ and $u_{b-1}u_b$. The vertex u_b has an index larger than a, so it differs from u_a and from u_{a-1}, whose indices are at most a. But u_b is an end of the second edge and is not an end of the first. So the two edges are different. □

In plain words. A cycle is a closed trip that never drives a road twice. The textbook lets such a trip pass a house twice on the way. A simple cycle is stricter. It passes no house twice, except the start, which is also the end. It is a ring of houses.

Definition 2.39 (cycle and simple cycle; textbook Definitions 1.2.14 and 1.2.15)**.** A closed trail is a *cycle* (the textbook also says *closed path*) [9, p. 22]. A cycle is a *simple cycle* if only its first and last vertices coincide [9, p. 22]; the textbook writes C_n for the simple cycle on n vertices, $n \geq 3$ [9, p. 22].

English usage differs: Grinberg's "cycle" is a closed walk of length at least 3 with distinct vertices except the closing one, which is the textbook's simple cycle [3]. This note follows the textbook and always writes "simple cycle" for the English "cycle". Cycles are not needed for the main results; they are given because the textbook's build order contains them.

Remark 2.40. A cycle may repeat vertices (a figure eight is a cycle) and may not repeat edges. The condition $n \geq 3$ excludes the degenerate cases: in a simple graph a closed trail cannot have length 1 or 2. The textbook does not say whether a closed trail of length 0 is a cycle; this note does not count it, as an interpretation.

Example 2.41. In T the sequence $1, 2, 3, 1$ is a simple cycle of length 3. In H the sequence v_1, v_3, v_4, v_2, v_1 is a simple cycle [9, p. 22]. The sequence $v_3, v_2, v_1, v_3, v_5, v_4, v_3$ in H is a cycle with six distinct edges that is not simple, because v_3 occurs three times [9, p. 22].

Non-example 2.42. In B the closed walk $1, 2, 1$ is not a cycle, because the edge n_1n_2 is used twice. In fact B has no cycle: this was checked by script for all closed walks of length 1 to 6 (Appendix D).

2.3 Distance

In plain words. The distance between two houses is the number of roads on the shortest trip between them. If there is no trip at all, we say the distance is infinite. A house is at distance zero from itself.

Definition 2.43 (distance; textbook Definition 1.2.16)**.** The *distance* $d(u, v)$ between vertices u and v is the length of a shortest (u, v)-path if at least one exists, and $+\infty$ otherwise [9, pp. 22–23].

The textbook's "path" in this definition is its Definition 1.2.12 (trail). The value does not depend on whether one reads trail or path, because the shortest walk, the shortest trail, and the shortest path between two vertices have the same length; this is proved in Appendix A.1. The textbook lists three properties [9, p. 23], proved from the definition in Appendix A.1:

1. $d(u, v) \geq 0$, and $d(u, v) = 0$ if and only if $u = v$;
2. $d(u, v) = d(v, u)$;
3. $d(u, v) \leq d(u, w) + d(w, v)$.

Remark 2.44. The definition allows the value $+\infty$, which is how a missing route shows up. It forbids negative values and any value that is not the length of an actual trip.

Example 2.45. In B: $d(n_1, n_2) = 1$, $d(n_2, n_3) = 1$, $d(n_1, n_3) = 2$, and $d(n_i, n_i) = 0$. The walk $1, 2, 1, 2, 3$ has length 4, but the distance from n_1 to n_3 is 2, the shortest.

Non-example 2.46. The number 4 is not the distance from n_1 to n_3 in B, although a walk of length 4 exists: distance is the minimum, not any length. In the graph D with vertices n_1, n_2, n_3 and the single edge n_1n_2, the distance $d(n_1, n_3)$ is not a number but $+\infty$.

2.4 Every walk contains a path

In plain words. Suppose you can drive from one house to another. Then you can do it without passing any house twice. Look at your trip. If it passes some house twice, you went round a loop. Cut the loop out. What is left is still a trip with the same start and the same end, and it is shorter. Keep cutting until no loop is left. The trip that remains is a route, and it is never longer than the first trip.

Lemma 2.47 (textbook Lemma 1.2.1, with the length bound)**.** *Let u and v be vertices of a simple graph, equal or different. If there is a (u, v)-walk of length k, then there is a (u, v)-path of length at most k, all of whose vertices and edges are vertices and edges of the walk.*

The textbook states the lemma for $u \neq v$ and without a length bound, and says that from any (u, v)-walk one can extract a simple (u, v)-path [9, p. 23]. Its proof is by induction on the length of the walk, cutting out the part between two equal vertices [9, p. 23], and that proof yields the bound. Grinberg's Corollary 2.9.7 states the version for all u, v with the length bound [3]. The case $u = v$ is needed here because a closed walk such as $2, 1, 2$ in B has equal ends; the path it contains is the single vertex, of length 0.

Proof. By strong induction on k we prove the statement $S(k)$: every (u, v)-walk of length k, for any u and v, contains a (u, v)-path of length at most k.

Base case $k = 0$. A walk of length 0 is a single vertex u, so $v = u$. A single vertex has no repeated vertex, so the walk is itself a path, of length $0 \leq 0$.

Induction hypothesis. Let $k \geq 1$ and assume $S(m)$ for every $m < k$.

Step. Let $u_0, \ldots, u_k$ be a (u, v)-walk, so $u_0 = u$ and $u_k = v$. If its vertices are pairwise distinct, it is a path of length k, and we are done. Otherwise there are indices $i < j$ with $u_i = u_j$. Form $W' = u_0, \ldots, u_i, u_{j+1}, \ldots, u_k$; if $j = k$ nothing follows u_i and W' ends at u_i. Then W' is a (u, v)-walk: it starts at $u_0 = u$; it ends at v, since its last vertex is $u_k = v$ when $j < k$ and is $u_i = u_j = u_k = v$ when $j = k$; and every two consecutive vertices are adjacent, since the only new consecutive pair is $(u_i, u_{j+1}) = (u_j, u_{j+1})$, which is a step of the old walk. The length of W' is $i + (k - j)$, which is less than k because $i < j$. By the induction hypothesis W' contains a (u, v)-path of length at most its own length, hence less than k. Every vertex and edge of that path belongs to W' and so to the old walk. □

Each cut removes at least one step, so the process stops after at most k cuts, and a walk that cannot be cut has no repeated vertex. As a trace, the textbook's walk $v_1, v_3, v_2, v_4, v_3, v_2, v_4, v_3, v_5$ of length 8 has $u_1 = u_4 = v_3$; cutting gives $v_1, v_3, v_2, v_4, v_3, v_5$ of length 5; cutting again gives v_1, v_3, v_5 of length 2, a path. In B, $1, 2, 1, 2, 3$ cuts to $1, 2, 3$, and the closed walk $2, 1, 2$ cuts to the single vertex 2.

In plain words. A route visits each house at most once. So a route cannot be longer than the number of houses minus one. That tells us how far we ever need to look for a route.

Corollary 2.48 (length of a path)**.** *In a graph with N vertices every path has at most $N - 1$ edges. Hence if there is a walk from u to v, there is a walk from u to v of length at most $N - 1$.*

Proof. A path of length k lists $k + 1$ pairwise distinct vertices, and there are only N vertices, so $k + 1 \leq N$. For the second part apply Lemma 2.47: it gives a path inside the walk, of length at most that of the walk and, by the first part, at most $N - 1$. □

Grinberg makes the same remark [3]. For B, $N = 3$, so a path has at most 2 edges, and the longest path in B has exactly 2.

2.5 Connected graphs and components

In plain words. Some villages are in one piece. From any house you can reach any other house along roads. Other villages are broken into pieces. In a broken village some house cannot be reached from some other house. Each piece, taken whole, is called a component.

Definition 2.49 (connected graph; textbook Definition 2.1.1)**.** A graph $G = (V, E)$ is *connected* if for any two vertices u and v of G there is a (u, v)-path in G [9, p. 48].

The board has the same definition: a graph is connected if for any n_i, n_j there exists a path between the nodes [4, 5].

Remark 2.50.

1. In the textbook the word "path" here is its Definition 1.2.12, a trail. The course convention is no repeated vertex, and the board does not say which it uses. The definition does not depend on the choice: a walk between two different vertices contains a path with distinct vertices (Lemma 2.47), and a walk with distinct vertices has distinct edges (Proposition 2.38), so the two readings accept the same graphs.

2. "Any two vertices" includes $u = v$, and for $u = v$ the walk with one vertex and no edge is a (u, u)-path. So these pairs are always satisfied, and reading the definition as "any two different vertices" changes nothing. The textbook does not discuss $u = v$, so this reading is an interpretation.

3. The definition forbids exactly one thing: a pair of vertices with no trip between them.

Example 2.51. B is connected. The pair n_1, n_3 is joined by the path $1, 2, 3$, and the pairs n_1, n_2 and n_2, n_3 by one edge each. The textbook's own examples are the complete graphs, the complete bipartite graphs, and three graphs of its Fig. 2.1.1 [9, p. 48].

Non-example 2.52. D, three vertices with only the edge n_1n_2, is not connected: no walk leaves n_3, so the pair n_1, n_3 has no path. The textbook shows a graph that is not connected and has four components in its Fig. 2.1.2 [9, p. 49].

In plain words. Look at a broken village. Say two houses belong together when one can reach the other. Houses that belong together form a piece. Nobody in one piece can reach anybody in another piece.

Definition 2.53 (connected components; textbook, p. 49, in prose)**.** On V define $u \sim v$ to mean that G has a (u, v)-path. This relation is reflexive, symmetric, and transitive, so V splits into classes $V_1, \ldots, V_p$. The subgraphs $G[V_j]$ with vertex set V_j and all edges of G that have both ends in V_j are the *connected components* of G [9, p. 49].

The textbook gives this definition in a paragraph without a number. The three properties of the relation hold because a single vertex is a path of length 0, a path can be read backwards, and two walks that meet at a vertex join into one walk, which contains a path by Lemma 2.47.

Remark 2.54. A component cannot be enlarged: a vertex reachable from a vertex of V_j lies in V_j. The textbook's Remark 2.1.1 says that G is connected if and only if it has exactly one component [9, p. 49].

Example 2.55. D has the two components $D[\{n_1, n_2\}]$ and $D[\{n_3\}]$; B has the single component $B[\{n_1, n_2, n_3\}]$.

Non-example 2.56. $D[\{n_1\}]$ is not a component of D, although it is a connected subgraph: n_2 can be reached from n_1, so n_2 belongs to the same class. A component is a whole class, not any connected piece.

3 Question 1: what the entries of A^k count

This section answers question 1. The answer is one sentence: the entry $(A^k)_{ij}$ is the number of walks of length exactly k from n_i to n_j (Theorem 3.1). The rest of the section proves it, draws the consequences that Section 4 needs, and shows that it counts walks and not paths.

3.1 The walk-counting theorem

In plain words. Take the table of roads and multiply it by itself. Each number in the new table counts trips of two roads. The trips run from the row house to the column house. Multiply again, and you count trips of three roads. The count includes trips that turn back. So the table counts trips, not clean routes.

Theorem 3.1 (walk counting)**.** *Let A be the adjacency matrix of a simple undirected graph G with vertices $n_1, \ldots, n_N$, and let $k \geq 1$ be an integer. Then $(A^k)_{ij}$ is the number of walks of length k from n_i to n_j, for all i, j.*

Quinlan's course notes state and prove this theorem for simple graphs [2, Theorem 1.1.3]; Grinberg states it for multidigraphs [3, Theorem 4.5.10], Lehman, Leighton, and Meyer for digraphs [6, Corollary 9.3.3], and Bai for simple graphs [1, Theorem 1]. The textbook does not state it [9], so it is proved here, along the route of Quinlan's proof.

Remark 3.2. The theorem allows $i = j$, which counts closed walks, and it counts every walk once, with repeats of vertices and edges permitted. It forbids reading the number as a count of paths (Proposition 3.6).

Example 3.3. For B, $(A^2)_{13} = 1$: the only walk of length 2 from n_1 to n_3 is $1, 2, 3$.

Non-example 3.4. $(A^2)_{12} = 0$ in B does not say that n_1 and n_2 are unconnected. It says only that no walk of length exactly 2 joins them; the walk $1, 2$ of length 1 and the walks $1, 2, 1, 2$ and $1, 2, 3, 2$ of length 3 do exist.

Proof of Theorem 3.1, by induction on k. Write $W_k(i, j)$ for the number of walks of length k from n_i to n_j. We prove $W_k(i, j) = (A^k)_{ij}$ for all i, j, for each $k \geq 1$. By Remark 2.23 we count vertex sequences $w_0, w_1, \ldots, w_k$ with $w_0 = n_i$, $w_k = n_j$, and w_{t-1}, w_t adjacent for each t.

Base case $k = 1$. A walk of length 1 from n_i to n_j is a sequence (n_i, n_j) with n_i, n_j adjacent. There is exactly one such sequence if n_i and n_j are adjacent and none otherwise. So $W_1(i, j) = A_{ij} = (A^1)_{ij}$.

Induction hypothesis. Let $k \geq 2$ and suppose $W_{k-1}(i, j) = (A^{k-1})_{ij}$ for all i, j.

Inductive step. Fix i and j. Every walk $w_0, \ldots, w_k$ of length k from n_i to n_j has a vertex w_{k-1} just before the end. Because $k \geq 2$, deleting the last vertex leaves a walk of length $k-1 \geq 1$, to which the hypothesis applies. Let $w_{k-1} = n_r$ with $r \in \{1, \ldots, N\}$ and sort the walks into N classes by the value of r. The classes are disjoint, since each walk has exactly one second-last vertex, and together they contain every walk, so

$$W_k(i, j) = \sum_{r=1}^{N} (\text{number of walks of length } k \text{ from } n_i \text{ to } n_j \text{ with second-last vertex } n_r).$$

Fix r. We claim that the number in class r is $W_{k-1}(i, r)\, A_{rj}$.

- If n_r and n_j are not adjacent, so $A_{rj} = 0$, no walk can have n_r followed by n_j, so class r is empty and its count is $0 = W_{k-1}(i, r) \cdot 0$.
- If n_r and n_j are adjacent, so $A_{rj} = 1$, delete the last vertex of a walk in class r. What remains is a sequence from n_i to n_r with consecutive vertices adjacent, that is, a walk of length $k - 1$ from n_i to n_r. Conversely, a walk of length $k - 1$ from n_i to n_r followed by n_j has all consecutive pairs adjacent (the new pair n_r, n_j is adjacent), so it is a walk of length k from n_i to n_j with second-last vertex n_r. Deleting and appending undo each other, so they match the two sets one to one, and the count is $W_{k-1}(i, r) = W_{k-1}(i, r) \cdot 1$.

Therefore $W_k(i, j) = \sum_r W_{k-1}(i, r) A_{rj}$. By the induction hypothesis $W_{k-1}(i, r) = (A^{k-1})_{ir}$, so

$$W_k(i, j) = \sum_{r=1}^{N} (A^{k-1})_{ir} A_{rj} = (A^{k-1}A)_{ij} = (A^k)_{ij}$$

by Definition 2.13. This proves the statement for k, and by induction for all $k \geq 1$. □

Why the argument is valid. The sum over the second-last vertex covers each walk exactly once, and the matching in the case $A_{rj} = 1$ is a one-to-one correspondence; these two facts are the only places where the counting could go wrong. Simplicity is used twice, as an interpretation of how the proof works: to identify a walk with its vertex sequence (Remark 2.23), and so that A_{rj}, which counts the edges between n_r and n_j, is 0 or 1. Undirectedness is not used in the induction; it only makes A symmetric, and the digraph versions of the theorem in the sources agree [3, 6].

3.2 Consequences

In plain words. A count is never negative, and it is always a whole number. The table of trips looks the same when flipped over its diagonal. A positive number means that at least one such trip exists.

Corollary 3.5. *Let $k \geq 1$.*

1. *Every entry of A^k is a non-negative integer, and a sum of powers of A has a positive (i, j) entry if and only if at least one summand has.*
2. *$(A^k)_{ij} > 0$ if and only if there is a walk of length exactly k from n_i to n_j.*
3. *A^k is symmetric.*

Proof. Parts 1 and 2: by Theorem 3.1 each entry is a count, and a sum of non-negative integers is positive exactly when some summand is. Part 3: $(A^k)^T = (A^T)^k = A^k$, because A is symmetric (Proposition 2.21). Alternatively, reversing a walk matches the walks from n_i to n_j with the walks from n_j to n_i. □

The diagonal of A^2 has a simple meaning, stated in Quinlan's notes after the theorem [2]. The walks of length 2 from n_i to itself are n_i, n_r, n_i with n_r a neighbour, one for each neighbour, so $(A^2)_{ii} = \deg(n_i)$. Hence the sum of the diagonal entries of A^2 is the sum of the degrees, which is $2|E|$ (Proposition 2.21). For B the diagonal of A^2 is $(1, 2, 1)$, the degrees, and its sum is $4 = 2|E|$ (Section 6).

3.3 Walks are not paths

In plain words. A trip may turn back and pass the same house twice. A route never passes a house twice. The table counts trips, so it counts the turn-back trips too. A house with one neighbour always has a two-road trip that goes out and returns. That trip is not a route.

Proposition 3.6 (the entry is not a count of paths or trails)**.** *In general $(A^k)_{ij}$ is not the number of paths, nor the number of trails, of length k from n_i to n_j.*

Proof. We use B. First, $(A^2)_{11} = 1 > 0$, the walk $1, 2, 1$. It repeats the vertex n_1, so it is not a path, and it repeats the edge n_1n_2, so it is not a trail. No path and no trail of length 2 from n_1 to n_1 exists, so the entry 1 is not a count of either. More generally, whenever n_i has a neighbour, $(A^2)_{ii} = \deg(n_i) > 0$ although a path of positive length cannot start and end at the same vertex. Second, $(A^3)_{12} = 2$, the walks $1, 2, 1, 2$ and $1, 2, 3, 2$; neither is a path or a trail, since each repeats a vertex and an edge. □

Table 2 shows the same thing for longer lengths. The number of walks from n_1 to n_2 grows with k (the matrices satisfy $A^{k+2} = 2A^k$ for $k = 1, \ldots, 5$, by script), while the only path from n_1 to n_2 is the single edge, of length 1.

Table 2: Walks and paths of length k from n_1 to n_2 in the board graph B, by enumeration.

k	walks from n_1 to n_2	of which paths
1	1	1
2	0	0
3	2	0
4	0	0
5	4	0

What a positive entry does guarantee is that a walk of length k exists, and then, by Lemma 2.47, that a path of length at most k exists. That is all that Section 4 uses. So the warning does not harm the connectivity test, but the value of an entry is not a path count.

A published statement that says paths. Paramadevan and Sotheeswaran state in their Theorem 1 that the entry of A^l counts the "different paths" of length l [7]. As written this is false for any meaning of "path" that forbids repeated vertices or repeated edges, by Proposition 3.6. The correct reading is walks, and their own property 7 says walks [7]. This note does not rely on that Theorem 1. Theorem 3.1 is a statement about a finite object and has also been checked by computer on all labelled simple graphs with $2, 3, 4, 5$ vertices (Appendix A.2); this check supports the proof and does not replace it.

4 Question 2: deciding connectivity from A

This section answers question 2. The answer is a test that uses only the matrix A: for $N \geq 2$, the graph is connected if and only if every off-diagonal entry of $A + A^2 + \cdots + A^{N-1}$ is positive (Theorem 4.7). The proof rests on two earlier results: walk counting (Theorem 3.1) and the lemma that every walk contains a path (Lemma 2.47).

4.1 The running total

In plain words. We need a name for a running total. It adds the counts of trips with one road, with two roads, and so on. We stop one step short of the number of houses. The whole check uses this one total.

Definition 4.1 (the sum S)**.** For a graph with $N \geq 2$ vertices let $S = A + A^2 + \cdots + A^{N-1}$, the sum of the first $N - 1$ powers of A, added entry by entry. This is notation introduced in this note and is not from the textbook.

Remark 4.2. By Theorem 3.1, S_{ij} is the number of walks from n_i to n_j whose length is between 1 and $N - 1$. Hence every entry of S is a non-negative integer. The diagonal of S counts closed walks and plays no role in the test (Remark 4.15).

Example 4.3. For B ($N = 3$), $S = A + A^2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}$, written out with the matrices in Section 6.

Non-example 4.4. For $N = 3$ the matrix A^2 alone is not S. In B, A^2 has the zero entry $(1, 2)$ although n_1 and n_2 are joined by an edge, and A alone has the zero entry $(1, 3)$ although n_1 and n_3 are connected. Either matrix alone would wrongly report "not connected".

4.2 The test and its proof

In plain words. We want a quick check for a village. We only have the table that says which houses share a road. From the table we can count trips. We count trips with one road, with two roads, and so on. We add the counts for each pair of houses. We never need trips longer than one road fewer than the number of houses. A wasteful trip can always be shortened. So longer trips tell us nothing new.

Proposition 4.5 (one pair at a time)**.** *Let G have $N \geq 2$ vertices and let $i \neq j$. The following are equivalent.*

(i) There is a path from n_i to n_j in G.

(ii) There is an integer k with $1 \leq k \leq N - 1$ and $(A^k)_{ij} > 0$.

(iii) $S_{ij} > 0$.

Proof. (i)⇒(ii). A path, in either reading of the word (Remark 2.50), is a walk, so there is a walk from n_i to n_j. By Lemma 2.47 there is a path P from n_i to n_j whose length k is at most that walk's length, and by Corollary 2.48 $k \leq N-1$. If k were 0, then P would be a single vertex and $n_i = n_j$; so $k \geq 1$. P is a walk of length k, so $(A^k)_{ij} \geq 1$ by Theorem 3.1.

(ii)⇒(iii). $S_{ij} = A_{ij} + (A^2)_{ij} + \cdots + (A^{N-1})_{ij}$ is a sum of non-negative integers (Corollary 3.5), so it is at least as large as the positive term from (ii).

(iii)⇒(ii). A sum of non-negative terms that is positive has a positive term $(A^k)_{ij}$ with $1 \leq k \leq N-1$.

(ii)⇒(i). If $(A^k)_{ij} > 0$ with $k \geq 1$, then by Corollary 3.5 there is a walk of length k from n_i to n_j. By Lemma 2.47 it contains a path from n_i to n_j with distinct vertices. A walk with distinct vertices has distinct edges (Proposition 2.38), so it is a path in either reading of the word. □

In plain words. There is a second form of the check. Let every house also stay where it is for one step. Then count again. It gives the same answer.

Proposition 4.6 (the form with $I + A$)**.** *Let G have $N \geq 2$ vertices, let $m \geq N-1$ be an integer, and let $i \neq j$. Then $\left((I+A)^m\right)_{ij} > 0$ if and only if $S_{ij} > 0$. Every diagonal entry of $(I+A)^m$ is at least* 1.

Proof. Since I commutes with A, expanding the product of m factors $I + A$ gives, by choosing I or A from each factor, the sum of $\binom{m}{k}A^k$ over $k = 0, \ldots, m$, because the product with k letters A is A^k and there are $\binom{m}{k}$ ways to choose which factors give A. Thus

$$(I+A)^m = \sum_{k=0}^{m} \binom{m}{k} A^k.$$

For $i \neq j$ the identity contributes 0 at position (i, j), so $\left((I+A)^m\right)_{ij} = \sum_{k=1}^{m} \binom{m}{k}(A^k)_{ij}$. Each coefficient $\binom{m}{k}$ with $1 \leq k \leq m$ is a positive integer and each $(A^k)_{ij}$ is a non-negative integer, so the sum is positive if and only if some $(A^k)_{ij}$ with $1 \leq k \leq m$ is positive. If such a $k \leq m$ exists, there is a walk, hence a path from n_i to n_j (Proposition 4.5), hence a positive $(A^{k'})_{ij}$ with $k' \leq N-1$ (same proposition), so $S_{ij} > 0$. Conversely $S_{ij} > 0$ gives a positive $(A^k)_{ij}$ with $k \leq N-1 \leq m$. On the diagonal the term $k = 0$ contributes $\binom{m}{0} I_{ii} = 1$ and all other terms are non-negative. □

In plain words. Here is the check in one sentence. Count trips of every length up to one fewer than the number of houses. Add the counts for each pair of different houses. If no pair has a total of zero, everyone can reach everyone.

Theorem 4.7 (the connectivity test)**.** *Let G be a graph with $N \geq 2$ vertices and adjacency matrix A. The following are equivalent.*

(a) G is connected.

(b) Every off-diagonal entry of $S = A + A^2 + \cdots + A^{N-1}$ is positive.

(c) Every entry of $(I+A)^{N-1}$ is positive.

Paramadevan and Sotheeswaran state (a)⇔(b) as their Theorem 2 [7], and MathWorld also states a matrix test for connectedness [11]; Bai states (a)⇔(c) as his Theorem 2 [1]. Their proofs say "paths" where walks are meant, and Paramadevan and Sotheeswaran pass from a non-zero entry to a path without the walk-to-path lemma, so these sources are cited for the statements only. The proof printed here is this note's own and follows the premises of Theorem 3.1 and Lemma 2.47.

Remark 4.8. The test allows any connected or disconnected simple undirected graph with at least two vertices. It forbids using only one power, or stopping before A^{N-1} (Section 4.3).

Proof. By Definition 2.49 and Remark 2.50, (a) says that for every pair of vertices there is a path between them, and for a pair with $u = v$ this is always true. So (a) holds if and only if statement (i) of Proposition 4.5 holds for every pair $i \neq j$. By Proposition 4.5 this is equivalent to (iii) for every $i \neq j$, which is (b). By Proposition 4.6 with $m = N - 1$, (iii) for every $i \neq j$ is equivalent to positivity of every off-diagonal entry of $(I + A)^{N-1}$, and the diagonal entries are positive anyway; together this is (c). □

In plain words. Here is the check step by step. First write the table of the village. Then multiply the table by itself again and again. After each step, add the new counts to the running total. Stop after one step fewer than the number of houses. Look at every pair of different houses. If any total is zero, those two houses cannot reach each other. If no total is zero, everyone can reach everyone.

Definition 4.9 (matrix test for connectivity)**.** *Input:* the adjacency matrix A of a graph with $N \geq 2$ vertices.

1. Set $P = A$ and $S = A$.
2. For $k = 2, 3, \ldots, N - 1$: replace P by PA (now $P = A^k$) and add P to S.
3. Look at the entries S_{ij} with $i \neq j$. If one of them is 0, report "not connected". If none is 0, report "connected".

The equivalent form computes $(I + A)^{N-1}$ and reports "connected" if no entry is 0.

Correctness is Theorem 4.7. A zero at (i, j) in step 3 says more than "not connected": by Corollary 4.13 below, n_i and n_j lie in different components.

Remark 4.10. The procedure allows any $N \geq 2$; for $N = 2$ the loop in step 2 is empty and $S = A$. It forbids reading the diagonal of S (Remark 4.15).

Example 4.11. For B and for D the procedure is carried out with every matrix written out in Example 6.2 and Example 6.3.

Non-example 4.12. Applying only step 1, that is, looking at A alone, is not the procedure: B is connected, but $A_{13} = 0$.

In plain words. The check can say more than yes or no. A zero in the total tells which two houses lie in different pieces. A nonzero total tells that two houses lie in the same piece.

Corollary 4.13 (components from S)**.** *For any graph, connected or not, with $N \geq 2$ vertices and $i \neq j$: $S_{ij} > 0$ if and only if n_i and n_j lie in the same component.*

Proof. Proposition 4.5 does not use connectivity of G: it says $S_{ij} > 0$ if and only if there is a path from n_i to n_j, and by Definition 2.53 this says that n_i and n_j are in the same class. □

Remark 4.14 (distance from powers). For $i \neq j$ in a connected graph, the smallest k with $(A^k)_{ij} > 0$ is the distance $d(n_i, n_j)$. Two sources state this [6, 7]. Here it follows from Corollary 3.5, which says that this smallest k is the length of a shortest walk, and from Proposition A.1, which says that the shortest walk has the length $d(n_i, n_j)$.

Remark 4.15 (why the diagonal is skipped, and the cases $N = 1$ and $N = 2$). For $N = 2$ and one edge, $S = A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has zeros on the diagonal although the graph is connected, so the diagonal of S cannot be required to be positive. For $N \geq 3$ and connected G the diagonal of S is positive anyway, because every vertex has a neighbour and $(A^2)_{ii} = \deg(n_i)$, but the test does not need this. For $N = 1$ there are no pairs of different vertices and the graph is connected; S is an empty sum and $(I + A)^0 = I = (1)$ has no zero entry, so the second form of the test gives the right answer, but the first form is not defined by the formula. Therefore the test is stated for $N \geq 2$.

4.3 Why $N - 1$ is the right last power

In plain words. Why stop at one road fewer than the number of houses? Because a route cannot be longer than that. Why not stop sooner? Take four houses in a row. To go from the first to the last, you must pass every house between. That takes three roads. If we counted only trips of up to two roads, we would miss this pair. We would say the village is cut in two, and we would be wrong.

Corollary 2.48 and the proof of Proposition 4.5 show that $N - 1$ is enough: if some walk joins two different vertices, then a path of length at most $N - 1$ does, so powers beyond A^{N-1} cannot change which off-diagonal entries of S are positive. The next proposition shows that $N - 1$ cannot be lowered.

Proposition 4.16 ($N - 1$ cannot be lowered to $N - 2$)**.** *For every $N \geq 3$ there is a connected graph with N vertices for which $A + \cdots + A^{N-2}$ has a zero off-diagonal entry.*

Proof. Take the path graph with edges $n_1n_2, n_2n_3, \ldots, n_{N-1}n_N$. Each step of a walk changes the vertex number by exactly 1. To go from vertex 1 to vertex N the number must increase by $N - 1$, so at least $N - 1$ steps are needed. A walk of length $k < N - 1$ from n_1 to n_N does not exist, so $(A^k)_{1N} = 0$ for $1 \leq k \leq N - 2$ by Theorem 3.1. Hence the $(1, N)$ entry of $A + \cdots + A^{N-2}$ is 0, although the graph is connected. □

In plain words. Here is the four-house village, with every table written out. The first table counts trips of up to two roads. It has a zero for the first and last house. The second table counts trips of up to three roads. It has no zero.

Example 4.17 (the bound is needed: the path on four vertices)**.** Let P_4 have vertices $n_1, \ldots, n_4$ and edges n_1n_2, n_2n_3, n_3n_4 ($N = 4$). Every product below was computed by script (Appendix D).

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \quad A^3 = \begin{pmatrix} 0 & 2 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 3 & 0 & 2 \\ 1 & 0 & 2 & 0 \end{pmatrix}.$$

$$A + A^2 = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} \ (N - 2 \text{ powers}), \qquad A + A^2 + A^3 = \begin{pmatrix} 1 & 3 & 1 & 1 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 1 & 1 & 3 & 1 \end{pmatrix} \ (N - 1 \text{ powers}).$$

With $N - 2 = 2$ powers the $(1, 4)$ entry is 0 and the test would wrongly say "not connected"; with $N - 1 = 3$ powers every entry is positive.

Table 3 confirms Proposition 4.16 for $N = 3, \ldots, 8$: the first power with a non-zero $(1, N)$ entry is $N - 1$ every time. For $N = 3$ the graph is B and the sum $A + \cdots + A^{N-2}$ is A itself, with $A_{13} = 0$.

Table 3: The path graph $n_1 n_2 \cdots n_N$: the first power with a non-zero $(1, N)$ entry, and the $(1, N)$ entry of the sums with $N-2$ and $N-1$ powers, by script.

N	smallest k with $(A^k)_{1N} > 0$	$(1, N)$ of $A + \cdots + A^{N-2}$	$(1, N)$ of $A + \cdots + A^{N-1}$
3	2	0	1
4	3	0	1
5	4	0	1
6	5	0	1
7	6	0	1
8	7	0	1

4.4 Breadth-first search, the alternative

In plain words. There is another way, and it needs no table of counts. Stand at one house and mark it. Then mark every house next to it. Then mark every house next to those. Keep going until no new house appears. If every house is marked, the village is in one piece.

Remark 4.18 (breadth-first search). Breadth-first search (BFS) from a start vertex s keeps a queue of discovered vertices, and repeatedly takes the vertex at the front and discovers its undiscovered neighbours. The correctness statement in the lecture slides of Schost is that for every vertex v there is a walk from s to v if and only if v is discovered when BFS ends [10]. Hence G is connected if and only if every vertex is discovered when BFS starts from any one vertex s [10], and BFS also yields the components directly. The matrix test and BFS agree on every labelled graph with at most 6 vertices (Section 6); this is a computer check and not a proof. The cost of the two methods is compared in Section 7.4.

5 Question 3: the definition of a path between n_i and n_j

Question 3 asks for the definition of a path between n_i and n_j. The answer is Definition 2.34 with the two ends named: *a path between n_i and n_j is a walk $w_0, w_1, \ldots, w_k$ with $w_0 = n_i$ and $w_k = n_j$ in which the vertices $w_0, \ldots, w_k$ are pairwise distinct; its length is k.* This is the textbook's simple path (Definition 1.2.13) [9, p. 22], and it is also what English usage calls a path [3]. For $i = j$ the only such path is the single vertex n_i, of length 0. That the course uses this meaning for "path" is stated by the assignment text; the textbook supplies the definition under the name "simple path". So the answer is an interpretation of the course convention, resting on that statement and on the textbook's definition, and it is qualified by the fact that the board itself does not say which meaning of "path" it uses [4, 5]. The rest of this section adds only what Definition 2.34 does not say: how the course word compares with the textbook's words and with English usage, and why the choice does not change which graphs are connected.

5.1 Three meanings of the word path

In plain words. Books disagree about the word path. Some use it for any trip that never drives a road twice. Others use it for a trip that never passes a house twice. So the same word can name two different kinds of trip. This note keeps one meaning for each word and writes the table below to show how the books line up.

Table 4 sets the three conventions side by side. It does not change any definition; it shows which word each source uses for each property of a walk.

The rows come from the textbook [9, pp. 21–22], from Grinberg's definitions of walk, path, and cycle [3], and from the Wikipedia article on paths, which records that some authors use "path" for walks with repeated vertices and "simple path" for distinct vertices, which is the

Table 4: The same properties of a walk under the textbook's words, English usage, and the words of this note. The numbers are the textbook's definition numbers.

Property of the walk	Textbook	English usage	This note
any sequence of steps along edges	walk (1.2.10)	walk	walk
first vertex equals last	closed walk (1.2.11)	closed walk	closed walk
no repeated edge; vertices may repeat	"path" (1.2.12)	trail	trail
no repeated vertex (so no repeated edge)	"simple path" (1.2.13)	path	**path**
closed, no repeated edge; vertices may repeat	cycle (1.2.14)	closed trail	cycle
closed, no repeated vertex except first = last, length at least 3	simple cycle (1.2.15)	cycle	simple cycle

textbook's pattern [8]. So the word "path" is used for two different properties: no repeated edge in the textbook's Definition 1.2.12, and no repeated vertex in English usage, where the textbook says "simple path". The course convention given by the assignment text is the second: no repeated vertex. For cycles the assignment text states no convention, so this note follows the textbook's word "cycle" for the closed trail and always writes "simple cycle" for the English "cycle".

5.2 The choice does not change connectivity

In plain words. You may worry that the two meanings give two different answers to the question whether a village is in one piece. They do not. If you can drive from one house to another at all, you can drive without passing a house twice. That trip also never drives a road twice. So both meanings accept the same villages.

Proposition 5.1 (the readings agree)**.** *For vertices u, v of a simple graph the following are equivalent: there is a (u, v)-walk; there is a (u, v)-trail; there is a (u, v)-path.*

Proof. A path is a trail (Proposition 2.38) and a trail is a walk, which gives path $\Rightarrow$ trail $\Rightarrow$ walk. If there is a walk, Lemma 2.47 gives a path. □

Grinberg's Proposition 2.9.10 states the equivalence of walks and paths only [3]; the trail step is the one proved here. So every statement of the form "there is a path between n_i and n_j" has the same truth value under the textbook's word and under the course's word, and Definition 2.49 and the test of Theorem 4.7 do not depend on the choice. What does depend on the choice is a count: the number of trails from u to v is not the number of paths, and neither is the number of walks (Proposition 3.6).

Example 5.2. In B there is exactly one path between n_1 and n_3, namely $1, 2, 3$, and between n_1 and n_1 there is only the path of length 0. In B every path is also a trail and every trail is a path of the same ends (by script, for lengths 0 to 4), because B has no cycle. To see a trail that is not a path, a graph with a cycle is needed, such as H: the trail $v_1, v_3, v_2, v_4, v_3, v_5$ is a trail but not a path [9, p. 22].

Non-example 5.3. In B the walk $1, 2, 1, 2, 3$ goes from n_1 to n_3 but is not a path between them, since n_1 and n_2 repeat. It is not a trail either, since the edge n_1n_2 repeats.

6 Worked examples

Every matrix in this section was produced by a script and is read into the text from a generated file, so that none was typed by hand (Appendix D). The example graph is B, the graph of the board matrix. The board drawing differs from the board matrix by one edge n_1n_3 and an arrowhead (Section 1.1); the matrix is followed.

6.1 The powers of the board matrix

In plain words. We take the village from the board. It has three houses in a row. The first and last houses each have one road. The middle house has two roads. We count trips of two roads, then trips of three roads. We also list every trip, so the reader can see that each count is right.

Example 6.1 (powers of the board matrix)**.** For B the matrix and its first powers are

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad A^3 = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix}.$$

Also $A^4 = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 2 \end{pmatrix}$ and $A^5 = \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 4 \\ 0 & 4 & 0 \end{pmatrix}$, and $A^3 = 2A$. The diagonal of A^2 is $(1, 2, 1)$, the degrees of n_1, n_2, n_3, and its sum is $4 = 2|E|$.

The walks behind these entries, with vertices written by index, are the following.

- Length 1: 1-2 and 2-3, and their reverses.
- Length 2: from n_1 to n_1 only 1-2-1 (one walk); from n_1 to n_3 only 1-2-3 (one); from n_2 to n_2 the walks 2-1-2 and 2-3-2 (two); from n_3 to n_3 only 3-2-3 (one).
- Length 3: from n_1 to n_2 the walks 1-2-1-2 and 1-2-3-2 (two); from n_2 to n_3 the walks 2-1-2-3 and 2-3-2-3 (two).

Each count equals the matrix entry in that position, and the script asserts this for every entry of A, A^2, and A^3. The sums of the induction step of Theorem 3.1 can be checked by hand:

$$(A^2)_{22} = A_{21}A_{12} + A_{22}A_{22} + A_{23}A_{32} = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 2,$$
$$(A^3)_{12} = (A^2)_{11}A_{12} + (A^2)_{12}A_{22} + (A^2)_{13}A_{32} = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 2.$$

In the second line the term for $r = 1$ is the length-2 walk 1-2-1 extended by the step to n_2, which gives 1-2-1-2, and the term for $r = 3$ is the walk 1-2-3 extended by the step to n_2, which gives 1-2-3-2.

In plain words. Now we sort the trips by kind. Some trips turn back and pass a house twice. Some trips drive a road twice. Only a few are clean routes. The table counts all trips of each length in the village of three houses.

Table 5 agrees with the matrices: the number of walks of length $k \geq 1$ is the sum of all entries of A^k ($4, 6, 8, 12$ for $k = 1, \ldots, 4$), and, for $k \geq 1$, the number of closed walks is the sum of the diagonal entries of A^k. It shows the gap between walks and paths: from length 3 on there are walks but no trails and no paths, because B has only 2 edges and a path has at most $N - 1 = 2$ edges (Corollary 2.48).

Table 5: Walks, trails, paths, and closed walks of each length k in B, counted over all start and end vertices by enumeration.

k	walks	trails	paths	closed walks
0	3	3	3	3
1	4	4	4	0
2	6	2	2	4
3	8	0	0	0
4	12	0	0	8

6.2 A connected example

In plain words. Take three houses in a row. A road joins the first and the second house. Another road joins the second and the third. The first and the third share no road. Can you still go from one to the other? Yes, through the middle house. We expect the check to say connected.

Example 6.2 (connected: the board matrix)**.** B has edges n_1n_2 and n_2n_3, $N = 3$, and $N-1 = 2$.

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad S = A + A^2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad (I + A)^2 = \begin{pmatrix} 2 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 2 \end{pmatrix}.$$

S has no zero entry, so by Theorem 4.7(b) G is connected, and $(I + A)^2$ has no zero entry, so by Theorem 4.7(c) it is connected: the two answers agree. The $(1, 3)$ entry of A is 0, so no road joins n_1 and n_3, yet $(A^2)_{13} = 1$: there is exactly one walk of length 2 from n_1 to n_3, namely 1-2-3. The $(1, 3)$ entry of $(I + A)^2$ is $1 = \binom{2}{2}(A^2)_{13}$, since the other terms of the expansion vanish there. The identity $(I + A)^2 = I + 2A + A^2$ holds entry by entry, by script. Breadth-first search from n_1 reaches $\{1, 2, 3\}$, which agrees.

6.3 A disconnected example

In plain words. Now take three houses again. Only one road exists, between the first and the second house. The third house stands alone. Nobody can walk to it. We expect the check to say not connected. It should also show that the third house is a piece by itself.

Example 6.3 (disconnected: one edge only)**.** D has the single edge n_1n_2, $N = 3$, and $N - 1 = 2$.

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad S = A + A^2 = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (I + A)^2 = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The off-diagonal zeros of S are at $(1, 3)$, $(2, 3)$, $(3, 1)$, and $(3, 2)$, and the zeros of $(I + A)^2$ are at the same four places. So by Theorem 4.7 the graph is not connected, and by Corollary 4.13 the vertices n_1 and n_2 lie in one component and n_3 in another: the components are $\{n_1, n_2\}$ and $\{n_3\}$, which agrees with breadth-first search. Higher powers do not help: the script checked that the entries $(1, 3)$ and $(2, 3)$ of A^k are 0 for $k = 1, \ldots, 6$, and in fact no walk reaches n_3 at all.

The example with $N - 1$ replaced by $N - 2$ is Example 4.17.

6.4 Agreement with breadth-first search on all small graphs

In plain words. We can also try the check on every possible small village. We compare its answer with the marking method. They agree every time. This does not replace the proof, but it would reveal a slip.

By computer, the test of Theorem 4.7(b), the form (c), the statement of Corollary 4.13, and the version of (c) with repeated squaring (Section 7.4) were compared with an independent breadth-first search on every labelled simple graph with $N = 2, \ldots, 6$ vertices. Table 6 gives the counts. In all $2 + 8 + 64 + 1024 + 32768$ graphs the four methods agreed with breadth-first search. The last column shows what goes wrong with $N - 2$ powers: it gives the wrong answer on exactly the connected graphs that are labelled paths, which number $N!/2$ for the sizes shown; for $N = 2$ the sum with zero powers is empty and is wrong on the one connected graph. This check covers $N \leq 6$ only.

Table 6: All labelled simple graphs with N vertices: how many there are, how many are connected, on how many the sum test gives the same answer as breadth-first search (all of them), and on how many the sum with only $N - 2$ powers gives the wrong answer.

N	graphs	connected	sum test agrees with search	wrong with $N - 2$ powers
2	2	1	2	1
3	8	4	8	3
4	64	38	64	12
5	1024	728	1024	60
6	32768	26704	32768	360

7 Discussion: limits, competing readings, and cost

7.1 What the results cover

The theorem and the test are proved for finite, simple, undirected graphs, the setting of the assignment and of the textbook's Definition 1.1.1 [9]. Where a loop or a repeated edge is allowed, an entry A_{rj} may exceed 1 or A_{ii} may be non-zero; the same induction then counts walks with multiplicity, but the items of Proposition 2.21 about entries 0 and 1, the zero diagonal, and the row sums change. This reading of the proof is an interpretation, in line with the versions of the theorem for directed graphs and multigraphs in the sources [3, 6]. Walk counting is also stated for those cases in the sources; the connectivity test was not examined for them, and the argument given here uses that A is symmetric and that a walk contains a path, so it should not be applied to directed graphs without further work. Weighted graphs and eigenvalue methods are outside this note.

The test says "connected" or "not connected". The sum S does not list a path, but Corollary 4.13 gives the components and Remark 4.14 the distances.

7.2 Conventions the note had to choose

- **The mark above A.** The board's mark in question 1 is not clearly a k (Section 1.1). The note reads it as a positive whole number k because the assignment text asks about A^k; if the mark is n or quotation marks, the statement of Theorem 3.1 still describes the entries for every positive exponent, including N [4, 5].

- **Drawing versus matrix.** The board drawing has an edge n_1n_3 and an arrowhead that the matrix does not have [4, 5]. The note follows the matrix because the problem says the graph is given through its adjacency matrix, and it does not guess which one the lecturer intended.

- **The word path.** The course meaning is stated by the assignment text and is the textbook's simple path; the board does not say which meaning it uses. Because of Proposition 5.1 the connectivity result is the same under the textbook's word.

- **Length zero.** The textbook does not say whether a walk of length 0 is allowed. The note allows it and does not count a closed trail of length 0 as a cycle (Remark 2.23); neither choice affects the main results, since a pair of different vertices needs a path of length at least 1.

No claim is made here about which answer to question 2 the lecturer intended: the board shows the question and not the answer. The note gives the sum test, the equivalent test with $I + A$, and breadth-first search, and proves the first two.

7.3 Competing explanations

Two sources that state the test explain it by counting "paths" [7, 1]. That explanation is wrong, because the powers of A count walks (Proposition 3.6); their statements of the test are correct and are cited as such. The second competing method is breadth-first search, which gives the same answer (Remark 4.18).

7.4 Cost of the two methods

In plain words. Which check is faster? The marking method looks at each house and road about once. The table method multiplies big tables many times. So the marking method is faster. The table method is still useful. Its answer is a clean statement about the table itself.

Remark 7.1 (cost; an estimate). Count scalar multiplications with the school method, which uses N^3 of them for one product of two N by N matrices. Reaching A^{N-1} from A takes $N-2$ products, that is $(N-2)N^3$ multiplications, which is 98,000,000 for $N = 100$. Because Proposition 4.6 holds with any exponent $m \geq N-1$ in place of $N-1$, one can instead square $I + A$ repeatedly until the exponent 2^t is at least $N-1$: this needs t squarings, $t = 7$ for $N = 100$, costing 7,000,000 multiplications. A script verified this squaring version against breadth-first search on every labelled graph with at most 6 vertices (Section 6.4). Breadth-first search takes time $O(n+m)$ for n vertices and m edges when the graph is given by adjacency lists [10]; when the input is the matrix A itself, it reads each row once, which is $N^2 = 10{,}000$ entries for $N = 100$. The entries of the powers also grow fast: for the complete graph on 10 vertices the largest entry of A^9 is 38,742,049 (script). Only the positions of the positive entries matter for the test, and because all entries are non-negative these positions depend only on which entries of the factors are positive, so a version that keeps just that information gives the same answer. Faster matrix multiplication exists and was not examined.

These numbers are operation counts under stated assumptions, not measurements; the comparison between the two methods is the note's own estimate, and the source of the breadth-first-search bound gives no comparison with the matrix test [10]. The matrix test is useful because its answer is a clean statement about powers of A, not because it is fast.

7.5 Checks that are not proofs

The walk-counting theorem was compared with a direct enumeration of walks on every labelled simple graph with $2, 3, 4, 5$ vertices, and the connectivity test with breadth-first search on every graph with up to 6 vertices (Appendix A.2). These finite checks support the proofs; they do not replace them.

8 Conclusion

Question 1. For a finite simple undirected graph with adjacency matrix A and an integer $k \geq 1$, the entry $(A^k)_{ij}$ is the number of walks of length exactly k from n_i to n_j (Theorem 3.1). It counts walks, in which vertices and edges may repeat, and not paths: in the board graph

$(A^2)_{11} = 1$ although no path of positive length starts and ends at n_1 (Proposition 3.6). The textbook does not state this theorem in the pages used, so it is proved in this note.

Question 2. A graph with $N \geq 2$ vertices is connected if and only if every off-diagonal entry of $A + A^2 + \cdots + A^{N-1}$ is positive, which holds if and only if $(I + A)^{N-1}$ has no zero entry (Theorem 4.7). The proof rests on walk counting and on the fact that a walk contains a path, which bounds the powers needed by $N - 1$ (Corollary 2.48); the bound cannot be lowered (Proposition 4.16). The test was shown on a connected and on a disconnected graph with three vertices, with every matrix written out (Example 6.2, Example 6.3). Breadth-first search is the alternative, and by the operation counts of the estimate in Remark 7.1 it needs far fewer operations (Remark 4.18).

Question 3. A path between n_i and n_j, in the course convention, is a walk from n_i to n_j in which no vertex occurs twice; this is the textbook's simple path, Definition 1.2.13 (Definition 2.34, Section 5). The textbook's own word "path" (Definition 1.2.12) means a walk with no repeated edge, which this note calls a trail. Whether the walk, trail, or path reading is used, a graph is connected under exactly the same conditions (Proposition 5.1).

What stays open. The board's mark above A is read as the exponent k on the strength of the assignment text; the board drawing and the board matrix differ by one edge, and the matrix is followed; and the textbook was read page by page only in the chapters used, so the absence of a matrix test in the rest of the book is reported with that qualification. None of these changes the mathematics proved above.

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[4] *Lecture whiteboard photo 1.* Photograph of the lecture board, file `sources/whiteboard-1.jpg`, supplied with the assignment. Viewed 2026-10-27. n.d.

[5] *Lecture whiteboard photo 2.* Second photograph of the same board at a slightly different angle, file `sources/whiteboard-2.jpg`, supplied with the assignment. Viewed 2026-10-27. n.d.

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A Secondary derivations

The main proofs are in the body: walk counting (Theorem 3.1), the walk-to-path lemma (Lemma 2.47), and the connectivity test (Proposition 4.5, Proposition 4.6, Theorem 4.7). This appendix proves the two statements about distance that the body uses, and describes the finite checks.

A.1 Distance: shortest walks, trails, and paths

In plain words. The distance between two houses was defined with the shortest trip that never drives a road twice, in the textbook's word, or the shortest route, in ours. We show that the shortest of every kind of trip has the same length. So it does not matter which word we use. Then we check the three facts about distance that the textbook lists.

Proposition A.1. *For vertices u, v of a simple graph the length of a shortest (u, v)-walk, of a shortest (u, v)-trail, and of a shortest (u, v)-path are equal. Each is $+\infty$ when the corresponding set is empty, and the three sets are empty together. Hence $d(u, v)$ in Definition 2.43 is the same whether "path" there is read as trail or as path.*

Proof. Every path is a trail (Proposition 2.38) and every trail is a walk, so the minimum over walks is at most the minimum over trails, which is at most the minimum over paths. By Lemma 2.47 a shortest walk contains a path of at most the same length, so the minimum over paths is at most the minimum over walks. Therefore the three minima are equal. If there is no walk, there is no trail and no path; if there is a walk, Lemma 2.47 gives a path. So the three sets are empty together. □

Grinberg makes the same remark: a shortest walk from u to v, if it exists, is also a shortest path [3].

Proposition A.2 (properties of distance)**.** *The three properties listed after Definition 2.43 follow from the definition.*

Proof. By Proposition A.1 we may compute d with shortest walks. *Property 1.* Lengths are $0, 1, 2, \ldots$ or $+\infty$, so $d \geq 0$. The walk consisting of u alone has length 0, so $d(u, u) = 0$. If

$d(u,v) = 0$ there is a walk of length 0 from u to v; it is a single vertex, so $v = u$. *Property 2.* If $u_0, \dots, u_k$ is a (u,v)-walk, then $u_k, \dots, u_0$ is a (v,u)-walk of the same length, since each step is an edge and edges are unordered. So the sets of walk lengths from u to v and from v to u are equal, and so are their minima. *Property 3.* If $d(u,w)$ or $d(w,v)$ is $+\infty$, the right side is $+\infty$ and the inequality holds. Otherwise take a shortest (u,w)-walk and a shortest (w,v)-walk and join them at w: the result is a (u,v)-walk of length $d(u,w) + d(w,v)$, and a shortest (u,v)-walk is no longer. □

A.2 Finite checks by computer

The following checks were run by the scripts of Appendix D. They are tests of statements already proved, not proofs.

- *Walk counting.* For every labelled simple graph with $N = 2, 3, 4, 5$ vertices (2, 8, 64, 1024 graphs), all walks were enumerated by brute force and compared with the entries of A^k ($k = 1, \dots, 5$ for $N \le 4$ and $k = 1, \dots, 3$ for $N = 5$): 40, 360, 5120, and 76800 entries respectively, all equal.
- *Connectivity test.* For every labelled simple graph with $N = 2, \dots, 6$ vertices (2, 8, 64, 1024, 32768 graphs, of which 1, 4, 38, 728, 26704 are connected) the sum test, the $(I + A)^{N-1}$ test, the component statement of Corollary 4.13, and the repeated-squaring version of Proposition 4.6 all agree with an independent breadth-first search (Table 6).
- *Walks, trails, paths.* The script for Section 2.2 classifies every example of that section by applying the definitions literally, and tests the construction of Lemma 2.47, Proposition 2.38, Corollary 2.48, and Proposition A.1 on all short walks of B, T, H and on random graphs; the saved output records no failure.

B Textbook locators and the board's notation

Table 7 lists, in the textbook's build order, where each concept used in this note is defined in the course textbook [9]. Page numbers are the printed ones; the PDF page is one higher. Only the pages named in the table, and the pages that carry the examples cited in the text, were read page by page; the formulas were checked on the page images.

The textbook's Lemma 1.2.1 is stated for two different vertices and without a length bound; Lemma 2.47 adds the case of equal ends and the bound, which Grinberg's Corollary 2.9.7 states [3].

Table 8 compares the notation of the board, of the textbook, and of this note.

The board's notation is read from the photographs [4, 5]; the difference is one of notation only.

C The board photographs

The two photographs show one board from slightly different angles, and they are read together [4, 5]. This appendix transcribes the board in reading order, reports what is clear and what is not, and compares the drawing with the matrix edge by edge.

C.1 Transcription

1. $A_{ij} = 1$ if $e_{ij} \in V$, and $A_{ij} = 0$ if $e_{ij} \notin V$. The word "if" in both lines is written on the board in Armenian and is transcribed here as "if". The board's V is the edge set.

Table 7: Where the course textbook defines each concept.

Concept	Textbook number	Page	Used in this note as
graph $G = (V, E)$	Definition 1.1.1	9	Definition 2.1
adjacent vertices	Definition 1.1.4	11	Definition 2.5
adjacency matrix $A(G)$	no number, after Definition 1.1.6	11	Definition 2.9
degree	Definition 1.2.1	16	Definition 2.17
sum of degrees is $2\|E\|$	Theorem 1.2.1 (Euler)	16	Proposition 2.21, item 5
walk, length	Definition 1.2.10	21–22	Definition 2.22
closed walk	Definition 1.2.11	22	Definition 2.26
walk with no repeated edge ("path")	Definition 1.2.12	22	Definition 2.30 (trail)
walk with no repeated vertex ("simple path")	Definition 1.2.13	22	Definition 2.34 (path)
cycle, simple cycle	Definitions 1.2.14, 1.2.15	22	Definition 2.39
distance	Definition 1.2.16	22–23	Definition 2.43
every walk contains a simple path	Lemma 1.2.1	23	Lemma 2.47
connected graph	Definition 2.1.1	48	Definition 2.49
connected components	prose, no number	49	Definition 2.53
connected if and only if one component	Remark 2.1.1	49	Remark 2.54

2. A drawing of three circled vertices labelled 1, 2, 3, beside a 3 by 3 table with rows 0 1 0, 1 0 1, 0 1 0 and the label "adj.".

3. "Problem. Given graph (G, V) with N nodes i.e. $|G| = N$ and the graph is represented through it's adj. matrix. So you have A matrix." The letter A here is written with a small mark on each side, which looks like quotation marks.

4. "1) What does A_{ij} represent", with a small mark above the A.

5. "2) How to know if graph is connected or not."

6. "Def. Graph is called connected if for any n_i, n_j $\exists$ path between nodes."

C.2 What is clear and what is not

Clearly legible: all the prose above, the matrix entries, and the vertex labels. *Likely reading:* the mark above A in question 1 is a power; the assignment text asks about A^k, so k is used. In both photographs the mark looks like a small n or a pair of quotation marks, and the board's own letter for the number of vertices is the capital N. *Unreadable:* a small scratched-out mark under vertex 1 in the drawing, which appears on both photographs and is not read.

C.3 Drawing against matrix

Table 9 compares the three possible edges of three vertices.

The matrix is symmetric with zero diagonal, so it is a valid adjacency matrix of a simple undirected graph (Proposition 2.21), and it does not support the arrowhead, which would mean

Table 8: Notation of the board, of the textbook, and of this note.

Object	Board	Textbook	This note
graph	(G, V)	$G = (V, E)$	$G = (V, E)$
edge set	V (in "$e_{ij} \in V$")	E	E
vertex	n_i	v_i	n_i
number of vertices	N, written $\|G\| = N$	n	N
edge between n_i, n_j	e_{ij}	uv	$n_i n_j$ or e_{ij}

Table 9: The board drawing against the board matrix, edge by edge.

Pair	Matrix	Drawing
$1, 2$	$A_{12} = A_{21} = 1$, an edge	a line, with an arrowhead at 2
$1, 3$	$A_{13} = A_{31} = 0$, no edge	a line
$2, 3$	$A_{23} = A_{32} = 1$, an edge	a line

a one-way edge. The matrix has two edges and the drawing has three; the line from 1 to 3 is in the drawing only. Following the problem statement, which says the graph is given through its adjacency matrix, this note uses the matrix, giving the graph B; the drawing read as drawn is the triangle T (Figure 1). The mismatch is reported and not resolved.

D Sources read and how to rerun the computations

D.1 Sources

The course textbook was read page by page at printed pages 9–12, 16, 21–23, and 48–49, with formulas checked on the page images; its text layer drops most symbols [9]. A full-text search of the whole book for the adjacency matrix found it on a few pages, and found no walk-counting theorem and no matrix test for connectivity; the pages outside those named were not read. Quinlan's notes were read in full [2], the other web and article sources in the parts that carry the statements cited [3, 6, 7, 1, 11, 8, 10], and both board photographs as images [4, 5]. All web sources were accessed on 2026-10-27.

D.2 Scripts

All scripts are in `appendices/scripts/`, use only the Python standard library and exact integer arithmetic, and are run from the paper folder.

- `m1_walks.py` (output `m1_walks_output.txt`): powers of the board matrix, enumeration of walks, trails, and paths, the hand-checkable sums, and the brute-force check of the walk-counting theorem.
- `m2_walks.py` (output `m2_walks_output.txt`): classification of the examples of Section 2.2 by the literal definitions, the walk-to-path construction and its checks, and distances.
- `m3_connectivity.py` (output `m3_connectivity.out.txt`): the connected and disconnected examples, the path graphs, the exhaustive comparison with breadth-first search, and the cost counts; the run takes about thirty seconds.
- `generate_paper_numbers.py` (output `generate_paper_numbers_output.txt`): writes the matrices and the table bodies that the text reads in from `figures/matrices.tex`; the tables of exhaustive and sharpness checks are read from the saved output of the previous script.

Each is run as `python3 -I appendices/scripts/<name>.py`. LaTeX is compiled with `latexmk -pdf` and `biber`.